SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
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Curriculum & Syllabi

Regulation 2022

2022-26 Batch

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

22AD009	AI IN HEALTH CARE APPLICATIONS	3/0/0/3
Nature of Course	F (Theory Programming)	
Prerequisite	Artificial Intelligence	

Course Objectives:

1	To Identify healthcare myths and digital transformation.
2	To gain knowledge in Precision Medicine and Intelligent Personal Health records.
3	To Understand AI Healthcare operations and Innovation.
4	To familiar with AIOps Strategy.
5	To analyse the future healthcare technologies.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C009.1	Understand about Health care myths and Digital Transformation.	[U]
C009.2	Recognize Health Records analytics.	[U]
C009.3	Identify the various healthcare operations.	[A]
C009.4	Develop an understanding in security services.	[A]
C009.5	Learn about telemedicine and their innovation.	[U]
C009.6	Apply principles and algorithms to evaluate a model.	[AP]

Course Contents:

MODULE I: INTRODUCTION **15 Hours**

AI health care myths - Human centered AI - Prescription for Personal Health - Ambient Computing Healthcare - Continuous monitoring using AI-Precision medicine -Intelligent Personal Health records - Digital Transformation- Case Study in Personalized Healthcare

MODULE II: AI HEALTHCARE OPERATIONS **15 Hours**

AIops strategy- Clinical Impact of AIops - Data Analytics and AI-Design and Innovation - AIops for Healthcare Delivery-AIOps for service performance - HIPAA, PH1, PII Protection - AIOps Usecase. Leveraging AIOps for Enhanced IT Operations

MODULE III: FUTURE OF HEALTHCARE **15 Hours**

Role of Medical Imaging Computing - AI in Radiology and Practical Use cases - Chronic Disease Management-AI Telemedicine - Telehealth Innovation-Digital Medication- Case Study: Cancer diagnostics and treatment decisions.

Total Hours:	**45**

Text Books:

1	Kerrie L. Holley, Siupo Becker, “AI -First Healthcare”, O'Reilly Media, Inc., 2018.
2	Adam Bohr, Kaveh Memarzadeh, “Artificial Intelligence in Healthcare”, Elsevier Science, 2020.
Reference Books:	
1	Robert Shimonski, Robert Shimonski”, “How Artificial Intelligence Is Changing IT Operations and Infrastructure Services” , wiley, 2020.
Web References:	
1	https://www.coursera.org/specializations/ai-healthcare
2	https://www.udemy.com/course/the-complete-healthcare-artificial-intelligence-course-2021